

Case Reference: CASE-ALT-DAF9C4
Transaction ID: TXN-1638CE92
Timestamp: 2026-08-16 15:22:26 UTC
Amount: **\$1,850.00 USD**
Action Taken: AUTO_BLOCK

RISK LEVEL: CRITICAL
96.6/100

Status: Transaction restricted; dossier preserved
for compliance and audit trail.

1. Multi-Factor Risk Model Diagnostics

Diagnostic Factor	Raw Metric	Factor Weight	Normalized Impact
Supervised Machine Learning (XGBoost)	99.3% Fraud Probability	65%	64.6 / 65.0
Unsupervised Anomaly (Isolation Forest)	85.0 Outlier Index	25%	21.2 / 25.0
Domain Heuristics & Rules	2 Rule(s) Triggered	10%	5.0 / 10.0
CALIBRATED COMPOSITE RISK SCORE	CRITICAL SEVERITY	100%	96.6 / 100.0

2. Explainable AI (SHAP) Forensic Feature Attribution

Primary Behavioral Triggers Identified:

- Elevated latent fraud index score (+4.56)
- Critical latent identity / card-use deviation (V14 = -4.29)
- Significant deviation on feature V4 (3.99)
- Critical latent identity / card-use deviation (V10 = -2.77)
- Critical latent identity / card-use deviation (V12 = -2.90)

Feature Name	Observed Value	SHAP Attribution (Delta)	Risk Direction
V14	-5.24	+0.3640	Elevates Fraud Probability
V12	-3.88	+0.2810	Elevates Fraud Probability
Amount_Deviation_Z	2.45	+0.1650	Elevates Fraud Probability

3. Fraud Analyst Investigation Notes & Audit Signature

Investigator: Lead Fraud Analyst #8142

Findings: Alert triage dossier.

Digital Audit Checksum (SHA-256):
6a9d0c811b0a3a516c551e30ff587687a1055d9d5baaaa2c076314406e910245

Verified Fraud Risk
Engine
COMPLIANCE CERTIFIED